

Towards Robust Models of Cyber-Physical Systems

Matthias Schaffeld

PhD Student

University of Duisburg-Essen

Duisburg, Germany

matthias.schaffeld@uni-due.de

Torben Weis

PhD Supervisor

University of Duisburg-Essen

Duisburg, Germany

torben.weis@uni-due.de

ABSTRACT

Cyber-Physical Systems (CPS) combine software with the physical world. For this purpose, CPS must model physical behavior in software. However, a software-based model cannot always accurately reflect the physical world. Often the model is a simplification of complex physical processes, or it suffers from measurement errors, or the physical side is subject to modifications and parameter drift, or the model is simply subject to misconceptions. It is an open research challenge how we can verify that physics and software-based model fit together. However, to rely on CPS in real-world scenarios we must ensure that physics and model are aligned. We propose a model formalism based on hidden Markov models that considers uncertainty and unknown phenomena and is robust enough to allow the analysis of CPS when working with error-prone data. More specifically, given observation data and an instance of the proposed model for a CPS (both of which may be flawed) the proposed formalism allows us to quantify the suitability between physics and model. If, however, a given model instance is deemed correct, the formalism enables methods which identify and smooth corrupt observation data as well as compute the most likely sequence of events for a given set of observations. Additionally, the formalism enables the learning of a suitable model according to given observation data. The model formalism will be tested with a simulation and a case study of an overhead traveling cargo crane system.

CCS CONCEPTS

• **Computer systems organization** → **Embedded and cyber-physical systems**; • **Computing methodologies** → **Modeling and simulation**.

KEYWORDS

cyber-physical systems, hidden Markov model, ubiquitous computing

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1 PROBLEM STATEMENT

From small-scale applications like wearable technology to large-scale applications in *Industry 4.0* there are plenty of systems that include both physical and digital systems as well as their mutual integration. Commonly, these hybrid systems use data of a physical system collected by sensors to estimate its current state and if necessary actuators to control it. However, this estimation is done by including some kind of digital model, representing the physical system [9].

Control engineering is an established approach to control physical processes, which makes use of control loops and differential equations to represent and regulate physical processes. It is intended to monitor certain process variables and keep them at a specified setpoint. Cyber-physical systems (CPS) on the other hand try to implement these hybrid systems by combining information and communication technology with physical systems. They are intended to enable real-time digital monitoring, analysis, and control of large and complex physical systems. Therefore, CPS require data of all important system aspects and a suited digital model of the physical system.

However, these requirements introduce a multitude of problems and questions about the longevity of CPS. First of all, it must be determinable if a proposed underlying model matches the physical system's reality. The model has to be able to accurately represent the physical system's behavior and determine its state by only perceiving data collected by sensors. This problem is further complicated by the fact that sensor data is prone to errors. Cheap or defective hardware causes bad measurements. Hardware failures or packet loss on the network introduce missing data. The preprocessing of data, for example with machine learning or deep learning, may result in misclassification. Furthermore, even if a model initially matches the physical system's reality, physical systems are exposed to external influences and consequently likely to change with time and wear. We cannot, therefore, assume that a once-suited model will remain so and have to identify these changes in case they are unwanted or unexpected. A suitable model therefore must be robust enough to handle errors, system changes and parameter drift or at least provide means to determine its progressive unsuitability.

2 RELATED WORK

As Lee [6] states CPS are about the intersection rather than the union of the cyber and the physical. They argue that CPS combine engineering models and methods from various fields of engineering and computer science, which do not combine easily and due

to the uncertain nature of physical systems do not preserve determinism. However, they further elaborate that models should not be conflated with their corresponding entity and introduce two deterministic CPS models, which ought to be practical and enable faithful physical realizations. Nevertheless, this research focuses on ensuring determinism regarding program execution and interaction with the physical world. It does not include uncertainties and nondeterminism coming from the physical system itself.

Zhang et al. [11] propose a conceptual CPS model to identify, define, and classify these uncertainties in CPS at the application, infrastructure, and integration level, which is then captured as a UML class diagram. This approach aims to provide a comprehensive description of expected uncertainties before deployment. It does however not provide any means of implementing a CPS that can handle these uncertainties, nor of analyzing system performance while dealing with them.

Broman et al. [2] provide an overview of viewpoints, formalisms, languages, and tools related to CPS design and propose a framework that considers the relevant viewpoints and assists in selecting a subset of the provided means to design a CPS. Their research focuses on the overall design of a CPS, whereby we are primarily interested in introducing a robust formalism to model the behavior of the physical system. However, the formalisms introduced are of interest and include state-machine-based formalisms, differential equations as well as timed and hybrid automata.

As defined by Henzinger [5], hybrid automata are used to model mixed discrete-continuous dynamic systems, such as a thermostat controlled by a microcontroller. The discrete microcontroller states are modeled as vertices of automata, which monitor specified variables, and the continuous dynamics of the heat are modeled by flow conditions, such as a differential equation. This approach goes a long way in capturing the continuous dynamics of physical systems. It does however only simulate the physical system and does not include observed data.

Other state-machine-based approaches are hidden Markov models (HMMs), which handle uncertainties by using transition probabilities between hidden, unobservable states and make use of actual observations by augmenting said states with emission probabilities of possible observations. Gosh et al. [4] use HMMs to model digital twins where the states and transition probabilities encapsulate the dynamics underlying the phenomenon and the actual phenomenon is recreated by using a Monte Carlo simulation process, which simulates observations. These simulated observations are then compared with a given time series. While this approach seems to include uncertainty through probabilities, real-world input and simulates physical behavior, it does not seem to account for errors like missing data.

Rocher et al. [8] introduce another model based on HMMs, that similar to hybrid automata includes a model of physical processes together with actual system inputs and outputs, namely the evidential input/output hidden Markov model (Ev-IOHMM). The Ev-IOHMM focuses on assessing the degree of effectiveness of a deployed CPS in the context of measure theory. More specifically it compares the expected simulated evolution of a CPS at run time with its concrete evolution to quantitatively assess the effectiveness with which the purpose of the CPS is met. This is a promising approach to establish benchmarks and monitor the effectiveness of CPS in

deployment, which should consequently improve the maintenance and longevity of these systems. However, as Rocher et al. state, the IOHMM cannot consider temporal constraints and is memory intensive. It remains to be seen, whether this approach can reasonably be used to analyze other aspects of CPS beyond their degree of effectiveness.

Wu et al. [10] also focus on the diagnostics and prognostics in CPS by focusing on performance degradation and run-to-failure states of a system by using a cluster-based hidden Markov Model (cHMM). Due to the uncertain nature of physical systems, they propose an unsupervised framework, to convert raw sensory streams into a sequence of observations, and learn hidden states and the dynamics of run-to-failure systems. This seems to be a promising approach to learn about possible performance degradation and failures of CPS. However, since it is an unsupervised framework we see possible problems by learning these aspects from the raw, error-prone, data. In summary, we found that many works attempt to infer system state from the data itself or to simulate and compare system behavior. However, we find this approach problematic if the source of the data is prone to error or changes over time and possibly without notice.

3 METHODOLOGY

We intend to develop a state-machine-based CPS formalism that can be used to model the assumed behavior of a CPS and establish a basis for analytic methods to solve the following problems. Given a model that is deemed feasible, for example by a domain expert, to reflect the behavior of a CPS, incorrect observations must be identified and corrected if possible. This specifically includes missing or incorrectly emerged data or events due to sensor and network errors. The same model should additionally be usable to calculate the most likely state sequences and expected observations. On the other hand, don't be given a feasible model, but rather observation data deemed feasible to reflect the behavior of a CPS, including observation errors, the suitability of models in question must be quantifiable. Additionally, the formalism should enable the learning of a suitable model according to the given observation data. Finally, if neither the model nor the observation data can be deemed feasible, the suitability between them must be quantifiable. However, we expect these problems to not constitute the entire problem space and hope to explore further application scenarios along the way. To fit the problems mentioned as well as possible, a formalism must be able to represent the uncertain and non-deterministic nature of CPS. By including probabilities, both probabilistic automata and HMM seem suitable for mapping these conditions. We do, however, lean towards HMM, as the introduction of hidden states fits the idea of unknown underlying phenomena in CPS. As defined by Rabiner [7], they are well suited to calculate the probabilities of observation sequences, the most likely accompanying state sequence, and estimate future behavior.

For HMM to apply to these problems, we must assume that a CPS is in one of N states at all times and the assumed underlying stochastic process needs to fulfill the Markov property. In addition, the associated probabilities must be time-independent and it must be possible to present observed data as discrete observations. It

remains to be seen whether such a formalism works with the assumption of a strong Markov property or whether higher-order Markov models are required to achieve better results. To require discrete observations should not pose a problem as such a formalism can be used after preprocessing on a more abstract level than raw streams of data. Furthermore, Wu et al. [10] have shown that cluster algorithms can be used to generate discrete observations from raw sensory stream data. These assumptions lead to a distance between the model and the real system, but we do not want to develop an all-encompassing CPS model, but one formalism that can serve as a reliable heuristic to address the above problems.

With these assumptions, an HMM could be used to estimate system behavior, calculate the probabilities of observation sequences, and determine erroneous behavior. However, this approach does not handle missing or incorrectly occurring observations, which are likely due to loss and retransmission of data. If sensor data is additionally preprocessed with machine learning or deep learning, the likelihood of false positives or false negatives increases as they cannot be eliminated. Therefore we want to expand the concept of HMM to include ϵ emissions (ϵ -HMM). These have been researched by Bahl et al. [1] as a text generator for speech recognition, but did not attract wide attention outside this context. However, in the context of CPS these can be used to include the possibility of missing observations.

Attempts to modify essential HMM algorithms, namely the Viterbi and the Baum-Welch algorithms, to include ϵ -emissions suggest that it is preferable to consider emissions to occur on state transitions – rather than on states, which would be usual for HMM. This also seems to be more convenient from the perspective of the system model design. HMMs with transitional emissions do not seem to have been formally defined, though they are mentioned by Callut [3] as a link between probabilistic automata and HMM. To combine the mentioned extensions to HMM and as a first attempt to construct a suitable formalism, we will define ϵ hidden Markov models with transition emissions (ϵ -HMMT).

After a suitable formalism has been found, means are needed to decide the stated problems. It seems essential to be able to quantify the suitability between a given model and observations, regardless if one is deemed feasible or not. If we can quantify the relationship between the two and one is deemed feasible, then the suitability of the relationship becomes the suitability of the other. However, there may be a better way to quantify the relationship, if one party is already deemed feasible. An approach to quantify the relationship could be to calculate an edit distance between the most likely sequence of observations and the actual sequence of observations. Another approach would be to divide the probability of the occurrence of an observed sequence by the probability of the occurrence of the most likely or a smoothed sequence. Both of these can be done and averaged over a large number of observation sequences. Given a sequence of observations, HMM are proficient in calculating the most likely sequence of hidden states that would cause this sequence. If an ϵ -HMMT can be modeled in such a way that each state corresponds to a possible observation, then the calculated state sequence to a given observed sequence corresponds to the most likely sequence of observations, that should have been observed without error. These sequences may of course match, but if not, probable errors can be found and the collected data can be

smoothed.

If the developed formalism and the accompanying methods are assessed as not being a reliable solution to the problems mentioned, it may be necessary to extend the formalism by incorporating the actual physical processes more closely. This could be done through means of measure theory or simulation on top of the ϵ -HMMT.

4 EVALUATION APPROACH

After its development, the proposed formalism and the accompanying methods will be evaluated using a simulation and data collected from an overhead traveling cargo crane system. The simulation of the crane system is currently used to train a model through reinforcement learning to handle the loading and unloading of containers as optimally as possible. It already uses a state machine, which allows us to capture the simulated system behavior by introducing discrete observations of different parts of the system. As a proof of concept, we will construct an ϵ -HMMT similar to this state machine in an effort to establish an initial benchmark of the model suitability. The next step will then be to compare the suitability of other models with this benchmark and to gradually introduce more and more errors into the system and observation sequences. The introduction of observation errors should consequently lead to a gradual decrease in suitability of models which do not take the possibility of errors into account, while we expect models based on the ϵ -HMMT approach to sustain higher levels of accuracy. In addition, we will learn and train models from the simulated data and see how they compare to the previously tested models. We will furthermore try to use different models to remove errors from the falsified observation sequences and investigate the relationship between *model suitability* and *success in error detection and elimination*.

If the simulated evaluation seems promising, the formalism and methods, as well as possible models that are suitable for the simulated system, are tested using a case study of an overhead crane system. The evaluation approach for the case study will be similar to the simulation. However, since its system is not embedded in a thoroughly controlled environment, we expect the case study to be a more reliable indicator of whether the developed formalism can be used as a means to enable analysis tools for the actual use of CPS.

5 EXPECTED CONTRIBUTION

The obstacles of modeling CPS arise from the heterogeneous nature of both networks and physical entities that are part of CPS and the inherent uncertainty of the underlying and complex physical systems [11]. We are therefore not aiming to develop an all-encompassing CPS model, but rather develop a formalism that allows accurate modeling of physical system behavior while being able to work with corrupt data and reliably estimate the physical system's state. Such a model should then be able to support the following three areas of application:

- Given a model deemed feasible, data quality can be enhanced by identifying and smoothing corrupt data as well as calculating the most likely sequence of observations and system states.

- Given data deemed feasible, the suitability of a given model can be quantified or a suitable model can be learned from the given data.
- Given both a model and data prone to errors, the suitability between them can be quantified.

If these three requirements are met, we will be able to generate a suitable CPS model, increase data quality in the short and long term, improve the analysis of CPS systems and by monitoring the suitability between data and model, identify system changes and degradation, and improve the longevity of CPS systems. Such a formalism cannot be assumed to infallibly reflect reality while dealing with a high degree of heterogeneity and uncertainty. However, it should serve as a heuristic to reliably address the problems mentioned and to cover the specified areas of application.

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BIOGRAPHICAL SKETCH

Matthias Schaffeld is a PhD student at the University of Duisburg-Essen and a research assistant at the university’s Chair of Distributed Systems in the Faculty of Engineering. His research focuses on the modeling of cyber-physical systems and the resulting analysis methods. His research advisor is Professor Torben Weis, who is also affiliated to the University of Duisburg-Essen as the head of the Chair of Distributed Systems in the Faculty of Engineering. Matthias Schaffeld started his PhD in September 2020 and is expected to be completed in September 2023.